NOISEMAP · DESIGN DECISIONS REGISTER

NoiseMap design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-02	Amish Chadha	Kit 1.9.1 upgrade; concept selection proposed (NSM-DDR-004); two fixes made under the fix rule (microphone, joint sealing); PR-003 and PR-002 proposals added
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 5 (NSM-DDR-003 accepted); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

NoiseMap design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Open decisions

Decisions below are proposed and not decided.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Which concept is carried forward	A pole clamp node; B balcony post node; C luminaire socket node; or A with B as a companion	Proposed, awaiting Amish: A (171) and B (163) are within 10 %, so no pick; C (152) not recommended	If A: nothing changes. If B or C: BOM, calculations and build plan for that option are new work	NSM-OPT-001, NSM-DDR-004

Decided under Amish's standing fix-rule instruction of 2026-10-23 (two fixes, both kept inside the working principle and recorded in NSM-STM-001): (1) the Infineon IM72D128 microphone is fitted by default in place of the ICS-43434, because the ICS-43434 clips at 113 dB SPL, which is only 103 to 108 dBA for traffic and is below the 110 dBA target (PR-004); the board is the dual-footprint one already decided (D3), so no part count changes; (2) every stainless screw, bolt and band that touches aluminium is sealed with neutral-cure silicone or fitted with a nylon washer, to limit crevice and galvanic corrosion in road salt (STM S15). Nothing that changes the working principle, pitch or scope was done.

Proposed, awaiting Amish (found by the independent STEM review, NSM-STM-001; they change targets, so they are not mine to make):

#	Decision needed	Options	Recommendation	Source
2	PR-003 (total uncertainty of 2 dB or less) is not met on paper: the budget gives 2.9 to 3.1 dB	Keep 2 dB and add a Class 1 reference at TRL 4; or relax the target to 3 dB (indicative-grade)	Relax to 3 dB for the indicative claim and keep 2 dB as a TRL 4 goal	NSM-CAL-001, NSM-STM-001 S14
3	PR-002 (fifteen 1-minute LAeq values per 15-minute record) cannot be sent at LoRaWAN SF10 to SF12 inside the duty-cycle and airtime limits	Keep PR-002 at SF7 to SF9 only; or accept longer intervals at slow rates	Accept the interval rule at SF10 to SF12 and state PR-002 and PR-012 as met by design at SF7 to SF9 only	NSM-CAL-001, NSM-STM-001

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The tube flange's three holes fall on a 45 mm circle, its base is 60 mm across or less, and its socket takes a 25 mm tube 25 mm deep	The saddle web holes and the band clearance (4 mm) assume it	NSM-DDR-003, P4
2	The processor board is no wider than 26 mm and no longer than 70 mm	It must fit the card guides printed in the head	NSM-DDR-003, P6
3	The windscreen is a 90 mm ball and takes a 40 mm bore, or comes bored to 40 mm	The push fit on the head and the head's position in the ball	NSM-DDR-003, P8
4	The band clamps close on poles from 60 to 140 mm, and the torque that gives 1,000 N of preload	The calculation note assumes that preload for every clamp margin	NSM-CAL-001, H6
5	The membrane's acoustic transparency and size from its maker's datasheet	It sits in the sound path; its effect is part of R5	NSM-DDR-001, D13
6	70 x 16 mm aluminium bar is stocked; if only 20 mm is, the blocks gain about 0.11 kg	The R10 margin is 0.05 kg	NSM-CAL-001, H2b
7	The FieldNode back plate as built matches FND-DWG-101 (V-block screw holes 18 mm each side, band slots 51 mm each side, at 20 and 270 mm up)	The street pole V-blocks use those holes and slots	NSM-DDR-003, P2
8	The membrane or neutral-cure silicone is compatible with the stainless-aluminium joints, and nylon washers fit under the M4 and M5 heads	The sealing fix of STM S15	NSM-STM-001, P4

Value engineering

Value-engineering target: USD 100 for the NoiseMap parts (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 72.00 (USD 28.00 under the target). The full node with the FieldNode core, which is costed in the FieldNode repository, is USD 211.00. Main cost drivers and savings worth trying:

- The largest NoiseMap lines are the arm and saddle with its flange and two bands (USD 16), the street pole adapter (USD 10), the sensor cable and gland (USD 9), the windscreen and spike, the level processor and the fixings (USD 8 each).
- Making the design constructable added USD 10 (two-band saddle and tube flange USD 6, fixings and lanyard USD 2, cable gland USD 1, head cap USD 1). The FieldNode core rose from USD 126 to USD 139 under its own design for construction.
- Savings worth trying: a printed or folded tube socket in place of the bought flange (about USD 3); one band size bought in a box for all four bands; an M12 cable with a moulded strain relief in place of the separate gland; the processor and microphone on one board once the head design settles (TRL 4 work).

Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D13: budget scope (NoiseMap parts only), privacy resting on open firmware, dual-footprint microphone, processing in the head, 15-minute records, shared calibrator, public notice, standard FieldNode core, microphone 4.0 m up and 0.45 m off the pole, A and C weighting, printed ASA head	Amish: "i accept all your recommendations, go with them across all repos."	NSM-DDR-001, NSM-DDR-002
2026-09-25	R10 relaxed to 3.5 kg with lighter V-blocks and saddle (O2); level-based wind flag (O3); interval rule at slow data rates (O4)	Amish, same instruction	NSM-DDR-002
2026-09-30	FieldNode core made constructable (the core NoiseMap now uses)	Amish: "i accept your recommended changes on design that are currently being sent across for my approval"	FieldNode FND-DDR-003
2026-09-30	Make every design physically buildable while drawing the build plan; keep open decisions out of the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	NSM-DDR-003 (changes accepted on 2026-10-02, below)
2026-10-01	`budget_usd` is a value-engineering target, not a limit	Amish: "the budgets are a hypothethical control target to ensure we are thinking along a value engineering lens"	This register, NSM-CAL-001 v0.3
2026-10-02	Design for construction accepted: the changes P1 to P11, as made	Amish: "i approve your recommendations for all 555 open decisions."	NSM-DDR-003
2026-10-02	Bird spike: the side spike in the skirt boss, 26 mm off the microphone axis, as modelled; the response is measured with and without it at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	NSM-DDR-003, A1
2026-10-02	R10 mass margin of 0.05 kg accepted; the prototype is weighed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	NSM-DDR-003, A2
2026-10-02	First co-design partner to approach: a city environmental noise team or a university acoustics group that can put a Class 1 reference sound level meter beside the node on the test street; a Sensor.Community group is a useful second contact for volunteer hosts	Amish: "i approve your recommendations for all 555 open decisions."	NSM-DDR-001, O1
2026-10-02	M12 sensor port pinout: ask FieldNode to adopt its own candidate pinout for both ports (pin 1 switched rail, pin 2 data A, pin 3 ground, pin 4 data B, pin 5 analog); NoiseMap uses pin 1 at 3.3 V, pin 3 ground and pins 2 and 4 for the UART pair	Amish: "i approve your recommendations for all 555 open decisions."	FieldNode FND-DDR-001, O2